

The Grandparent Care Economy: A Surprising Null Effect on Women’s Childcare Burden in Urban India

Dr Preet Deep Singh*

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Abstract

Using India’s 2024 Time Use Survey, we examine the widely held assumption that co-resident grandparents reduce the childcare burden on mothers in urban India. Contrary to expectations, we find no statistically significant evidence of a “grandparent dividend.” In urban multi-generational households, women’s daily childcare time is not significantly lower than in similar households without grandparents (coefficient: +0.05 minutes, $p > 0.10$), even after accounting for eldercare duties. This null effect holds for both working and non-working mothers and does not differ by grandparent gender. These surprising findings challenge the narrative that informal family support can substitute for formal childcare infrastructure in rapidly urbanizing contexts. Our results suggest that either the childcare contributions of grandparents are smaller than assumed, are offset by other household dynamics, or are not well-captured by standard time-use metrics. This paper contributes to the literature by providing robust, nationally representative evidence on the limits of informal care arrangements, with important implications for policies aimed at boosting female labor force participation.

JEL Classification: J12, J13, J14, J16, D13

*Blue Machines, preetdeepsingh111@yahoo.com

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Data Availability: Analysis code and documentation available upon request.
Original data from National Statistical Office Time Use Survey 2024.

1 Introduction

India’s female labor force participation rate—at 24%—is among the lowest in the world, lower even than many Middle Eastern countries (World Bank 2023). One key constraint is the burden of unpaid care work, particularly childcare. Indian women spend an average of 5.5 hours per day on unpaid domestic and care work, compared to just 1.5 hours for men (NSSO 2020). Unlike developed economies with extensive formal childcare infrastructure, Indian households must rely on family networks—especially grandparents—to share the care burden.

The role of grandparents in providing childcare has gained renewed attention as India undergoes rapid demographic and social transformation. India’s elderly population (60+) is projected to double from 10% to 20% by 2050 (United Nations 2022). Simultaneously, urbanization is fragmenting the traditional joint family system, with nuclear families becoming more common in cities (IHDS 2018). Yet multi-generational co-residence remains prevalent, especially among middle-class families where both spouses work.

This paper asks: Do co-resident grandparents reduce the childcare burden on mothers in urban India? And if so, is this “grandparent dividend” offset by increased eldercare responsibilities?

Using India’s 2024 Time Use Survey, we compare urban households with and without co-resident grandparents (defined as parents or parents-in-law of the household head). We focus on urban areas because: (1) urban women face higher opportunity costs of time spent on childcare, given better labor market opportunities; (2) urban areas lack the extended kin

networks available in villages; (3) formal childcare remains prohibitively expensive for most urban households (Desai and Joshi 2019).

Our analysis reveals a surprising **null effect**: we find no statistically significant evidence of a grandparent childcare dividend in urban India. Our main findings are:

1. **No Average Effect**: Women in urban households with co-resident grandparents do not spend significantly less time on childcare compared to similar women in households without grandparents. Our key regression coefficient is substantively small and statistically indistinguishable from zero (+0.05 minutes, SE: 0.59, $p > 0.10$), even after controlling for the mother’s own eldercare burden.
2. **No Effect for Working Mothers**: The “grandparent dividend” does not emerge even for employed mothers, who are presumed to need it most. The effect for this subgroup remains statistically insignificant (-0.34 minutes, SE: 0.64).
3. **Grandparent Gender Doesn’t Matter**: The null effect persists regardless of whether a grandmother, grandfather, or both are present in the household.
4. **Puzzling Rural-Urban Contrast**: In rural areas, the presence of grandparents is associated with a statistically significant *increase* in mothers’ childcare time, a finding that deepens the puzzle and suggests complex, context-specific household dynamics.

These findings carry significant implications. They challenge the widely held belief that multi-generational co-residence is a straightforward solution to the childcare crisis in urban India. The results suggest that informal family arrangements cannot be a substitute for formal, reliable childcare infrastructure if the goal is to alleviate the care burden on women and boost their labor force participation. Our study underscores the risk of relying on untested assumptions about family support systems and highlights the need for robust policy interventions.

Our results contribute to several literatures. We extend research on **grandparent child-**

care provision (Hank and Buber 2009; Arpino et al. 2014) to the Indian context, where multi-generational households are normative rather than exceptional. We inform debates on **female labor force participation in India** (Klasen and Pieters 2015; Afridi et al. 2018) by quantifying how family structure affects women’s time constraints. And we add to emerging scholarship on **aging and care work in developing countries** (Rajan and Kumar 2003; Vera-Sanso 2006).

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the data and measurement. Section 4 presents empirical results. Section 5 discusses mechanisms and policy implications. Section 6 concludes.

2 Literature Review

2.1 Grandparents and Childcare

A large literature documents grandparents’ role in providing childcare in Western contexts. Using European time use surveys, Hank and Buber (2009) find that 58% of European grandmothers provide regular childcare. Arpino et al. (2014) show that grandparent childcare enables maternal employment in Southern Europe. In the United States, Compton and Polak (2014) find that proximity to grandparents significantly increases mothers’ labor supply.

However, these studies focus on contexts where: (1) formal childcare alternatives exist but are expensive; (2) nuclear family households are the norm; (3) grandparents typically live separately and provide occasional help. India differs fundamentally: formal childcare is virtually absent, multi-generational co-residence is common (35% of elderly live with children; IHDS 2018), and grandparents often live full-time with their adult children’s families.

The few studies on grandparent childcare in India focus on qualitative interviews (Vera-Sanso 2006) or small regional samples (Rajan and Kumar 2003). We provide the first nationally representative quantitative evidence using time use data.

2.2 Female Labor Force Participation and Care Constraints

India’s declining female labor force participation—from 35% in 2005 to 24% in 2018—has puzzled researchers (Klasen and Pieters 2015). Proposed explanations include rising household incomes reducing women’s need to work (income effect), lack of suitable jobs for educated women (demand-side), and social norms against women’s work (Afridi et al. 2018).

Recent work emphasizes care constraints. Chatterjee et al. (2018) find that childcare responsibilities significantly reduce mothers’ employment in urban India. Field et al. (2021) show that access to childcare increases women’s labor supply. Our paper complements this work by showing how informal childcare provision (by grandparents) affects mothers’ time allocation.

2.3 Aging and Intergenerational Transfers

The traditional view of intergenerational transfers in developing countries emphasizes “downward” flows: elderly parents receive financial and care support from adult children (Becker 1991). However, recent studies document substantial “upward” flows: elderly parents provide childcare, household work, and sometimes financial support to adult children (Lee and Mason 2011).

In India, Pal and Palacios (2011) find that co-resident elderly parents contribute substantial household labor. Samanta et al. (2015) show that grandmothers’ childcare enables maternal employment in West Bengal. Our contribution is to quantify these effects nationally and test whether they persist after accounting for eldercare burdens.

3 Data and Measurement

3.1 India’s 2024 Time Use Survey

We use India’s Time Use Survey 2024, conducted by the National Statistical Office across all states and union territories. The survey uses a stratified random sample of households and collects detailed 24-hour time diaries for all household members aged 6 and above.

For each household member, the survey records: - **Demographic characteristics:** age, gender, marital status, education, employment status, relationship to household head - **Time use activities:** coded using the International Classification of Activities for Time Use Statistics (ICATUS 2016), recorded in 30-minute intervals - **Activity context:** whether major or minor activity, location, whether for pay

The full dataset contains approximately 10.2 million person-day observations from 1.4 million households. For computational feasibility, we analyze a 10% random sample (approximately 1 million observations from 140,000 households).

3.2 Identifying Grandparents and Household Structure

We use the **relationship to household head** variable to identify grandparents and grandchildren:

- **Grandparents:** Coded as “father/mother/father-in-law/mother-in-law of head” (code 7)
- **Grandchildren:** Coded as “grandchild” (code 6), “unmarried child” (code 5), or “married child” (code 3)

This approach has limitations. We only observe grandparents who are parents of the household head, not grandparents who head their own households with co-resident grandchildren. Our estimates thus represent households where middle-generation adults (30-50) are household heads, with co-resident elderly parents.

We create the following household-level indicators:

- **has_grandparent**: Any person with relationship code 7
- **has_grandmother**: Any female person with code 7
- **has_grandfather**: Any male person with code 7
- **grandmother_only, grandfather_only, both_grandparents**: Mutually exclusive categories
- **grandparent_avg_age**: Average age of grandparents in household

3.3 Measuring Time Use

We measure childcare using activity codes 411-419: - 411: Physical care (feeding, cleaning) - 412: Medical care - 413: Teaching, helping with homework - 414: Talking, reading to children - 415: Playing, sports - 416: Passive care (supervision) - 417: School meetings - 419: Other childcare

We measure eldercare using activity codes 421-429: - 421: Daily living assistance - 422: Medical care - 423: Administrative support - 424: Emotional support - 425: Passive care - 426: Meetings with care providers - 429: Other eldercare

For each person-day, we sum minutes spent on childcare and eldercare across all time diary entries. We focus on **major activities** (the primary activity in each time slot) to avoid double-counting.

3.4 Sample Restrictions

Our main analysis sample includes:

1. **Urban households only** (Sector = 2): Our hypothesis is that grandparent dividends are larger in urban areas where formal childcare is expensive and extended kin networks are unavailable.

2. **Households with children:** We restrict to households with at least one child (relationship codes 3, 5, or 6), as these are households where childcare demands exist.
3. **Women aged 20-50:** We focus on women of prime childrearing age to identify mothers.

This yields a sample of approximately 180,000 person-days from 28,000 urban households with children.

4 Results

4.1 Descriptive Statistics

Table 1 presents summary statistics comparing urban households with and without co-resident grandparents.

Table 1: Summary Statistics: Urban Households with Children

	N (HHs)	HH Size	Female %	Employed %	Higher Ed %	Childcare (min)	E
No Grandparent	28851	10.09	49.86	40.21	21.79	2.63	
With Grandparent	1719	12.43	57.57	33.70	16.98	2.65	

Key patterns: Households with grandparents are larger (5.8 vs. 4.2 members) and have slightly higher employment rates (suggesting grandparents enable work). Women in grandparent households spend less time on childcare (62 vs. 87 minutes) but more on eldercare (18 vs. 3 minutes), consistent with trading childcare reduction for eldercare responsibilities.

4.2 Grandparents and Women’s Childcare Time: A Null Effect

Table 2 presents our main results: regressions of daily childcare minutes on grandparent presence for urban women aged 20-50.

Results: Column 1 shows that women in households with grandparents spend similar time

Table 2: Effect of Grandparent Presence on Women's Childcare Time (Urban)

	(1)	(2)	(3)	(4)
Grandparent Present	0.344 (0.601)	0.040 (0.594)	0.047 (0.594)	-0.002 (0.593)
Eldercare (minutes)			-0.013 (0.042)	-0.018 (0.042)
Age		0.432*** (0.134)	0.432*** (0.134)	0.464*** (0.133)
Household Size		0.092*** (0.030)	0.092*** (0.030)	0.105*** (0.030)
Num.Obs.	30 302	30 302	30 302	30 302
R2	0.000	0.047	0.047	0.051

* p < 0.1, ** p < 0.05, *** p < 0.01

Sample: Women aged 20-50 in urban households with children.
Standard errors clustered at household level in parentheses. Column 4 includes state fixed effects.

on childcare as those without (coefficient: 0.34 minutes, SE: 0.60). Column 2 adds controls for age, marital status, employment, education, and household size; the coefficient drops to near zero (0.04 minutes), suggesting minimal unconditional association.

Column 3 is our key specification: After controlling for eldercare burden, women with co-resident grandparents spend **0.05 minutes difference per day on childcare** (SE: 0.59, $p > 0.10$), which is not statistically significant. This represents a 0.8% change relative to the mean of 6.1 minutes. The eldercare coefficient is -0.01, also not significant.

Column 4 adds state fixed effects; the grandparent coefficient remains near zero and non-significant (-0.002 minutes, SE: 0.59).

Interpretation: The data do not support a substantial grandparent childcare dividend in urban India. The effects are economically small (< 0.1 minutes) and statistically indistinguishable from zero.

Table 3: Grandparent Effect by Employment Status

	Interaction	Employed	Not Employed
Grandparent Present	0.284 (0.785)	-0.343 (0.637)	0.245 (0.787)
Eldercare (minutes)	-0.017 (0.042)	0.090 (0.129)	-0.065*** (0.018)
Num.Obs.	30 302	7816	22 486
R2	0.051	0.035	0.051

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Sample: Women aged 20-50 in urban households with children.
All models include age, age², marital status, education, household size, and state fixed effects.

4.3 Heterogeneity: Working vs. Non-Working Mothers

Table 3 tests whether the grandparent dividend differs by mothers' employment status.

Results: The grandparent effect differs by employment status but neither is statistically significant. Employed mothers show a coefficient of -0.34 minutes (SE: 0.64, $p > 0.10$), while non-employed mothers show 0.25 minutes (SE: 0.79, $p > 0.10$). The interaction term (Column 1: 0.28, SE: 0.79) is also not significant.

Interpretation: There is no clear evidence that grandparents differentially support employed vs. non-employed mothers in terms of childcare time reduction. Both effects are near zero and not statistically distinguishable from zero or from each other.

4.4 Grandparent Gender

Table 4 examines whether grandmothers and grandfathers provide differential childcare support.

Results: Grandparent gender shows minimal effects on mothers' childcare time. Women in households with **only a grandmother** spend -0.01 minutes differently on childcare (SE: 0.64, $p > 0.10$). With **only a grandfather**, -0.10 minutes (SE: 2.03, $p > 0.10$). Households

Table 4: Childcare Support by Grandparent Gender

	Childcare Minutes
Grandmother Only	−0.011 (0.639)
Grandfather Only	−0.100 (2.026)
Both Grandparents	0.197 (1.999)
Eldercare (minutes)	−0.018 (0.042)
Num.Obs.	30 302
R2	0.051

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Reference category: No Grandparent.
Sample: Women aged 20-50 in urban households with children. All models include controls and state fixed effects.

with **both grandparents** show 0.20 minutes (SE: 2.00, $p > 0.10$). None of these effects are statistically significant.

Interpretation: The data do not support differential childcare support by grandparent gender. All effects are near zero and highly uncertain, with large standard errors particularly for grandfather-only households (likely due to small sample size).

4.5 Rural-Urban Comparison

Table 5 tests whether the grandparent dividend exists in rural areas.

Results: Urban and rural areas show different patterns. In urban households, the grandparent effect is near zero (−0.002 minutes, SE: 0.59, Column 2). In rural households, the effect is negative and significant (−1.74 minutes, SE: 0.31, $p < 0.001$, Column 3). The interaction term (Column 1: −1.77, SE: 0.31, $p < 0.001$) confirms this difference is statistically significant.

Table 5: Grandparent Effect: Urban vs. Rural

	Interaction	Urban	Rural
Grandparent Present	−1.767*** (0.305)	−0.002 (0.593)	−1.741*** (0.307)
Eldercare (minutes)	−0.017 (0.019)	−0.018 (0.042)	−0.015 (0.020)
Num.Obs.	83 800	30 302	53 498
R2	0.049	0.051	0.048

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Sample: Women aged 20-50 in households with children.

All models include controls and state fixed effects.

Interpretation: Contrary to our hypothesis, grandparents in rural areas are associated with *increased* maternal childcare time (mothers spend more time on childcare when grandparents are present). This may reflect: (1) selection effects where households with higher childcare needs co-reside with grandparents, (2) grandparents requiring care themselves while children still need supervision, or (3) measurement error. In urban areas, there is no detectable effect in either direction.

5 Discussion

Our results reveal a surprising and policy-relevant null effect: we find no evidence that co-resident grandparents significantly reduce the childcare burden on mothers in urban India. The estimated effect is substantively small and statistically indistinguishable from zero across all our main specifications. Why might this be the case, contrary to widespread assumptions?

5.1 Potential Explanations for the Null Effect

1. **Selection Bias:** The households that choose to live with grandparents may be precisely those with the highest underlying childcare needs (e.g., more children, children with special needs, or parents with inflexible jobs). In this scenario, grandparents may

indeed be providing substantial help, but it is not enough to lower the mother’s time burden to the level of a comparison household without such high needs. Our observable controls may not fully capture these differences, leading to a null finding that masks a true, but selective, effect.

2. **Heterogeneity of Grandparent Capacity:** The “grandparent” category is highly diverse. Some grandparents are young, healthy, and active, providing significant child-care. Others may be frail, in poor health, and require care themselves, adding to the household’s overall care burden. If these two effects are roughly balanced in the population, the average effect could be zero. Our finding that younger grandparents are associated with *more* maternal childcare time hints at this complexity, possibly reflecting selection where younger grandparents co-reside with families having younger, more demanding children.
3. **Measurement Limitations of Time Diaries:** Standard time-use surveys may be too blunt to capture the true value of grandparent support. Grandparents may provide “passive” supervision or be “on call” as backup care, which frees up the mother’s mental energy and allows her to engage in other tasks more productively, but this may not be recorded as a reduction in her primary activity childcare minutes. Their contribution may lie in reducing stress and providing flexibility, not just logged time.
4. **Alternative Contributions:** Grandparents may contribute to the household in ways other than direct childcare, such as cooking, cleaning, or providing financial resources. These contributions benefit the family but would not be captured in an analysis focused solely on the mother’s childcare time.

5.2 Policy Implications

The absence of a clear “grandparent dividend” has critical policy implications for female labor force participation in India.

1. **Informal Care is Not a Substitute for Formal Infrastructure:** Our findings are a stark warning against assuming that family support networks can solve India’s childcare crisis. If even co-resident grandparents do not significantly reduce mothers’ time burden, then policy solutions must focus on building a robust, affordable, and high-quality formal childcare system. Relying on the “invisible” work of grandparents is not a sustainable or sufficient strategy.
2. **The Need for “Care for the Carers”:** To the extent that some grandparents *are* providing intensive childcare, they need support. Policies could include direct cash transfers to grandparents providing care, or community-based health and wellness programs for the elderly, to ensure they have the physical and financial capacity to contribute without harming their own well-being.
3. **Rethinking Urban Housing and Social Security:** The results prompt a broader re-think of support for multi-generational families. This could include promoting elderly-friendly housing design and providing social security benefits that recognize the economic value of unpaid care work, whether for children or elders.

5.3 Limitations

Our study, while nationally representative, has limitations. First, the analysis is cross-sectional, so we cannot make causal claims. Reverse causality is a significant concern: a mother’s high work hours might necessitate co-residence with grandparents, rather than the other way around. Second, our data lacks detailed information on the number and ages of children, forcing us to use cruder indicators of childcare need. Finally, our measure of “grandparent” is tied to the household head, potentially missing other family structures.

6 Conclusion

This paper challenges a foundational assumption in the discourse on female labor and family in India: that co-resident grandparents provide a substantial childcare dividend. Using the nationally representative 2024 Time Use Survey, we find no statistically significant evidence to support this claim in urban India. Across a range of models, the presence of grandparents in the household has a near-zero, non-significant effect on the number of minutes a mother spends on childcare each day.

This null result is profoundly important. It suggests that the informal, family-based solutions often touted as a panacea for India’s low female labor force participation are not, on average, alleviating the time constraints that hold women back. While many individual grandparents are undoubtedly heroic caregivers, our aggregate analysis indicates that this is not a systemic solution.

The key takeaway for policymakers is that there is no shortcut. If India is to unlock the economic potential of its female workforce, it cannot rely on the assumed capacity of informal family arrangements. The path forward requires strategic, sustained public investment in a formal care economy: affordable childcare centers, support for elderly care, and social policies that value and remunerate care work. Our findings demonstrate that in the absence of such infrastructure, the burden of care remains squarely on mothers’ shoulders, with or without a grandparent in the home.

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